



Research paper

Enhancing real-time IoT intrusion detection using KAN-based frameworks with SMOTE

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ABSTRACT

The rapid proliferation of Real-Time Internet of Things (RT-IoT) applications has introduced increasingly complex security challenges, making Intrusion Detection (ID) a critical component for maintaining system integrity and operational reliability. Traditional Intrusion Detection Systems (IDS) often struggle to effectively manage the high-dimensional, dynamic, and heterogeneous nature of IoT network traffic, which can include diverse attack types and large volumes of data. This study investigates three variants of the Kolmogorov–Arnold Network (KAN) for intrusion detection using the RT-IoT2022 dataset. First, a Simple-KAN network is implemented as a lightweight solution suitable for resource-constrained IoT devices, providing rapid learning and low computational overhead. Second, a Transformer-KAN model is evaluated, leveraging attention mechanisms to capture sequential dependencies and temporal patterns in network traffic. Third, a GKAN is developed to model device-to-device relationships and network interactions through graph representations. To address class imbalance and enhance minority class representation, both the Synthetic Minority Oversampling Technique (SMOTE) and class weighting are applied across all models. Experimental evaluation demonstrates that Simple-KAN with SMOTE achieves the highest test accuracy at 99.62%, closely followed by Transformer-KAN with SMOTE at 99.55%, indicating consistent and highly effective detection across diverse attack types. In contrast, GKAN with class weighting achieves 99.39%, exhibiting comparatively lower and more variable performance. Data balancing provides modest improvements for Simple-KAN and Transformer-KAN and is particularly beneficial for GKAN, underscoring the need for tailored strategies in graph-structured models. These results confirm that KAN-based architectures, especially Simple-KAN and Transformer-KAN, offer a promising and practical approach for enhancing IoT security in real-time, multi-class intrusion detection scenarios.

1. Introduction

The RT-IoT market is growing rapidly across several critical infrastructures. In healthcare, for example, the market stood at approximately \$44.21 billion in 2023 and is anticipated to reach close to \$170 billion by 2030, driven by technologies such as remote patient monitoring and wearables that enhance real-time care and efficiency (Yassen et al., 2023a; Hussein et al., 2023). In addition, industrial applications that benefit from RT-IoT with higher automation and predictive maintenance, which equate to higher productivity and less downtime. In the case of Unmanned aerial vehicles (UAVs), for instance, IoT sensors are employed for environmental monitoring and disaster response applications, among others, with the ability to provide timely data and decision-making (Ahmed et al., 2025; Mohammed et al., 2023).

Given the critical roles performed by RT-IoT devices, any compromise of their integrity or availability may result in devastating consequences, such as threats to human safety, financial losses, and degradation of essential services. The highly connected and dynamic nature of RT-IoT environments render them especially susceptible to sophisticated cyber-attacks, which may prey on network or device-inherent vulnerabilities (Yassen et al., 2023b). Thus, having a robust and efficient IDS, designed to deal with the differentiating characteristics of RT-IoT, is imperative to protect these infrastructures and ensure their reliable operation (Fortune Business Insights, 2025). However, ID in RT-IoT systems presents unique and complex challenges. The diverse and dynamic nature of RT-IoT traffic patterns, driven by heterogeneous devices, varying communication protocols, and fluctuating data rates,

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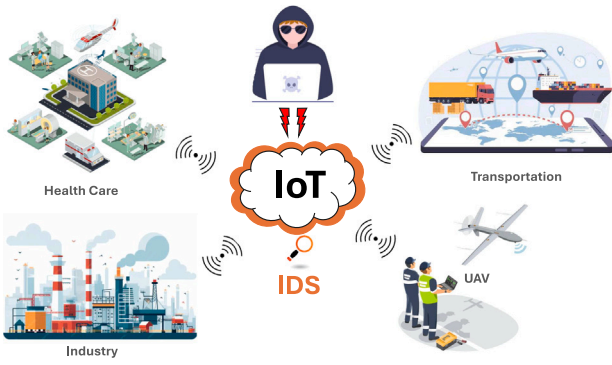


Fig. 1. IoT application domains and intrusion risks.

makes it difficult to establish consistent baselines for normal behavior. This complexity increases the risk of both false positives and false negatives in detection systems (Fares et al., 2025; Ghorbani et al., 2025). Furthermore, RT-IoT environments often operate under strict constraints on computational resources, power consumption, and latency, especially in safety-critical applications. Consequently, there is a pressing need for ID models that achieve a balance between detection accuracy and computational efficiency, enabling timely and reliable threat identification without compromising the real-time performance of the underlying system (Mohammed et al., 2024). Fig. 1 illustrates the most important IoT application domains and intrusion risks.

Artificial neural networks (ANNs) have long been applied in IDS across various domains, including RT-IoT environments (Kikissagbe and Adda, 2024). Traditional neural architectures such as Multi-Layer Perceptrons (MLPs) and Convolutional Neural Networks (CNNs) (Sworna et al., 2023; Shen et al., 2024) are effective, but face challenges related to model size, interpretability, and computational efficiency (Chinnasamy et al., 2025; Jiang et al., 2025). These challenges have motivated the exploration of alternative architecture, among which Kolmogorov–Arnold Networks (KANs) have recently garnered significant attention (Ji et al., 2025; Yu et al., 2024b).

KANs are inspired by the Kolmogorov–Arnold representation theorem and distinguish themselves from conventional MLPs in a fundamental way as shown in Fig. 2: they employ learnable univariate functions on the edges (connections) between neurons, rather than fixed activation functions at nodes (Liu et al., 2025; Nuaimi et al., 2023). Each weight parameter is replaced by a spline-parameterized function, granting the network greater flexibility to be adapted to complex data patterns. This design allows KANs to achieve improved accuracy with fewer parameters, making them computationally efficient and more interpretable (Somvanshi et al., 2025).

The advantages of KANs extend beyond parameter efficiency. Their edge-based activation flexibility supports faster neural scaling laws, enabling the networks to generalize better as they grow in size (Mohan et al., 2024). KANs can be pruned and sparsified to remove redundant functions, further enhancing interpretability without sacrificing performance. This ability to uncover symbolic mathematical forms through visualization has positioned KANs as valuable collaborators in scientific discovery, bridging AI with theoretical research.

In the specific realm of intrusion detection, KANs enhance intrusion detection by replacing conventional MLP layers in CNNs, leading to more accurate and parameter-efficient models (Z. Wang et al., 2025). Recent studies have shown that KAN-based CNNs provided superior classification performance on benchmark datasets and reduced model's size (Koenig et al., 2024). Furthermore, hybrid IDS approaches that integrate KANs with ensemble methods further boost detection accuracy, making them well-suited for the evolving nature of IoT network traffic (Kich et al., 2024; Kheddar et al., 2023). In addition, KAN-Mixer architectures and convolutional variants showed stronger resilience against adversarial perturbations compared to baseline MLP

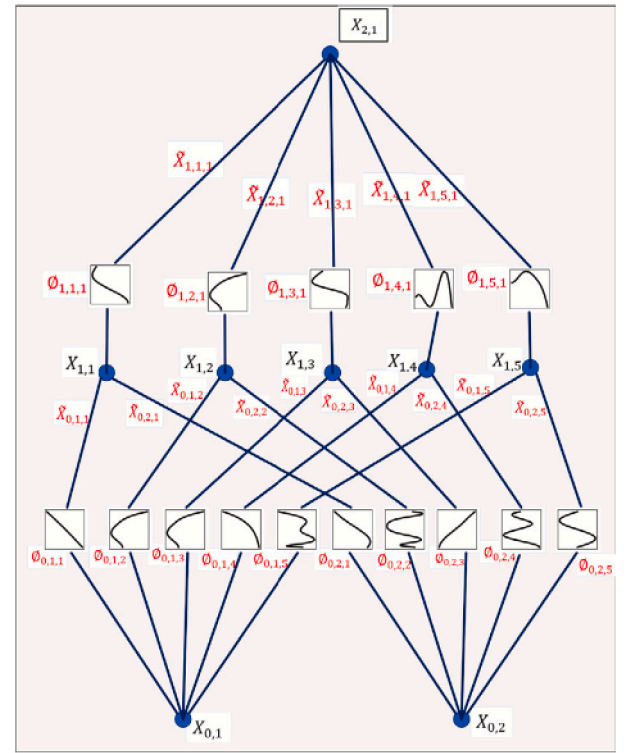


Fig. 2. The KAN architecture proposed by Liu et al. (2025).

models (Ibrahim et al., 2024). This robustness is essential for securing IDS implementations against evasion tactics and malicious manipulation. Challenges remain around optimizing training efficiency, energy consumption, and memory usage, but ongoing research continues to advance these fronts, positioning KANs as a transformative architecture in machine learning and security domains (Tongtao Liu, 2025).

This study advances real-time ID by comparing three KAN-based architectures: Transformer-KAN, Simple-KAN, and GKAN. Transformer-KAN leverages attention mechanisms to extract critical sequential features from network traffic, achieving strong detection accuracy in complex IoT environments. Simple-KAN offers a lightweight design that reduces computational overhead while maintaining reasonable performance, serving as a baseline for evaluating more advanced models. GKAN incorporates graph-based representations to capture relational patterns among IoT devices, making it particularly effective in distributed networks. In addition to training the models on the publicly available RT-IoT2022 dataset, we also explored strategies to mitigate class imbalance. Specifically, we employed the SMOTE (Chawla et al., 2002) to generate additional synthetic samples for minority classes and applied class weighting to modify the loss function (Chamseddine et al., 2022). These approaches ensured that underrepresented classes received greater attention during training and helped the models achieve more balanced performance. The main contributions of this work can be summarized as follows:

- Three KAN-based architectures, Transformer-KAN, Simple-KAN, and GKAN are explored for real-time intrusion detection, with the goal of identifying the best-performing model.
- Implementation of multi-class real-time intrusion detection on the RT-IoT2022 dataset, with the integration of SMOTE and class weighting to effectively mitigate class imbalance across all evaluated models.
- Comprehensive comparison with state-of-the-art methods, demonstrating that Simple-KAN achieves superior results in terms of detection accuracy and robustness.

The paper is structured as follows. Section 2 reviews recent advances in intrusion detection and KAN-based approaches. Section 3 outlines the overall methodology adopted in this study. Section 4 presents the used datasets, including RT-IoT2022, as well as the implementation details and evaluation criteria. Section 5 provides a detailed analysis of the models, covering loss and accuracy trends, confusion matrices, violin plot visualizations, qualitative assessments, and comparative evaluations with state-of-the-art methods. Finally, Section 6 summarizes the findings and emphasizes the superior performance of Transformer-KAN in multi-class real-time intrusion detection.

2. Related works

Recent advances in IoT ID have leveraged various DL architectures to address the challenges of multi-class attack classification, class imbalance, and real-time constraints. This section reviews both conventional deep learning approaches and emerging KAN-based methods, summarized in Table 1.

2.1. DL approaches for ID

CNNs have been widely adopted for spatial feature extraction in network traffic analysis. Recent work by Alsubaei (2025) proposed an optimized Sequential Neural Network (OSNN) with comprehensive preprocessing and hyperparameter tuning via Grid Search, achieving strong multi-class intrusion detection performance across varied datasets. However, CNN-based approaches typically require large parameter counts and face challenges in capturing temporal dependencies inherent in network traffic patterns.

Recurrent architectures address temporal modeling limitations by learning sequential dependencies. Sinha et al. (2025) presented an enhanced LSTM-CNN hybrid architecture for real-time IoT intrusion detection, achieving 90.2% accuracy under adversarial attack conditions with a false positive rate of 0.13%. Similarly, Gueriani et al. (2024) proposed a CNN-LSTM model evaluated on CICIoT2023 and CICIDS2017 datasets, achieving 98.42% accuracy with an F1-score of 98.57%. While effective, these architectures often suffer from computational overhead and gradient vanishing problems in long sequences.

Recent research has explored combining multiple architectural components to leverage complementary strengths. Ghosh et al. (2025) introduced TACNet, a temporal attention-based CNN framework combining multi-scale feature extraction with LSTM networks and attention mechanisms for adaptive feature focusing in IoT/IIoT environments. Yang et al. (2024) proposed an attention-enhanced BiGRU-Inception-CNN model for Industrial IoT intrusion detection, demonstrating that attention mechanisms can selectively focus on critical features while reducing computational complexity.

Recent studies have applied Deep Reinforcement Learning (DRL) to IoT intrusion detection to address dynamic environments, zero-day attacks, and limited labeled data. Yu et al. (2024b) proposed DC-IDS, which models open-set intrusion detection as a Markov decision process using DQN and conditional VAEs to achieve fine-grained classification of known traffic and recognition of unknown attacks. Shen et al. (2024) introduced DQN-HIDS, an LSTM-based DQN framework for SIoT that improves autonomous labeling under insufficient training samples and outperforms traditional methods. Yu et al. (2024a) further modeled attacker-detector interactions as adversarial stochastic games using DDQN with lazy penalties to balance detection accuracy and efficiency. Although these DRL-based approaches demonstrate adaptability and reduced reliance on labeled data, they are computationally intensive, face real-time limitations on resource-constrained edge devices, and mainly support binary or coarse-grained classification, restricting their applicability to fine-grained multi-class intrusion detection in heterogeneous IoT environments.

2.2. KAN-based approaches for ID

Recent research, has highlighted the potential of KANs to enhance neural network architectures and intrusion detection systems by improving representational capacity, robustness, and adaptability.

In Amouri et al. (2024), a hybrid intrusion detection framework was introduced that combines KANs with XGBoost, achieving detection accuracies above 99% and precision, recall, and F1-scores exceeding 98%. This model demonstrated superior performance compared to conventional MLP-based networks in handling complex and dynamic IoT environments, capitalizing on the representational flexibility of KANs and the robust classification capability of XGBoost to enhance both accuracy and interpretability.

Convolutional Kolmogorov–Arnold Networks (CKANs) (Abd Elaziz et al., 2024) proposed KAN layers as a replacement for the conventional MLP layers within CNN architectures. CKANs have been demonstrated to achieve superior performance on IDS tasks, particularly in IoT environments. Evaluated on benchmark datasets outperform established deep learning models including CNNs, RNNs, and Autoencoders in both binary and multi-class intrusion detection, achieving accuracies up to 99.93% with fewer parameters. Despite these gains, CKANs face limitations in memory consumption, execution speed, and energy efficiency, identifying areas for future optimization (Z. Wang et al., 2025).

A recent work proposed the Transformer with Kolmogorov, Arnold Networks (TFKAN) (Fares et al., 2025) for IoT intrusion detection by replacing the standard MLP layers with KAN layers. Unlike MLPs, KANs use learnable univariate function components, enabling more compact representations with fewer trainable parameters. Evaluations on RT-IoT2022, IoT23, and CICIoT2023 datasets achieved accuracy scores of 99.96%, 98.43%, and 99.27%, respectively. On average, the TFKAN model is 78% lighter than MLP-based Transformers, highlighting the potential of KANs as an efficient alternative in IDS for IoT environments.

GraphKAN (Wu et al., 2025), a combination of Graph Attention Networks (GAT) with KAN, has been proposed as an intrusion detection framework for smart grids. Unlike traditional GNN methods that mainly use network data, GraphKAN integrates both physical and logical device dependencies to better capture infrastructure interactions. The GAT module extracts global features, while the KAN enhances nonlinear representation through learnable activation functions. Experiments on datasets from Mississippi State University and Oak Ridge National Laboratory achieved detection accuracies of 97.63%, 98.66%, and 99.04% for binary, ternary, and 37-class tasks, respectively.

2.3. Research gap and contributions

In summary, conventional deep learning approaches (CNN, LSTM, Bi-LSTM, attention mechanisms) achieve strong performance (97%–98% accuracy) but often require large parameter counts (780K–5.3M parameters), suffer from interpretability limitations, and face computational efficiency challenges in resource-constrained IoT environments. While recent KAN-based approaches demonstrate superior parameter efficiency and comparable or better accuracy (99+%), there exists no comprehensive comparison of various KAN architectures under the prevalent class imbalance problem observed in IDS datasets.

To fill this gap, this paper systematically explores three KAN-based architectures, Simple-KAN, Transformer-KAN, and GKAN, while adopting SMOTE and class weighting to evaluate the superior methodology for real-time multi-class IoT intrusion detection. Our work provides the first comparative analysis of KAN architectural variants specifically addressing class imbalance, offering practical insights for deploying efficient and interpretable IDS in resource-constrained IoT environments.

Table 1
Summary of recent DL and KAN-based intrusion detection approaches for IoT.

Study	Technique type	Datasets	Key results/Limitations
Gueriani et al. (2024)	CNN–LSTM	CICIoT2023, CICIDS2017	98.42% accuracy, F1 98.57%; high training cost
Yang et al. (2024)	Attention-based CNN–RNN	IIoT datasets	High accuracy with reduced complexity; still parameter intensive
Amouri et al. (2024)	KAN + ML hybrid	IoT benchmark datasets	Accuracy >99%; improved interpretability and robustness
Abd Elaziz et al. (2024)	KAN-based CNN	Multiple IoT datasets	Accuracy up to 99.93%; memory and energy overhead
Yu et al. (2024b)	DRL (DQN + VAE)	TON-IoT	Detects unknown attacks; computationally expensive and limited real-time suitability
Shen et al. (2024)	DRL (LSTM–DQN)	UNSW-NB15	Reduces manual labeling; mainly binary/coarse-grained detection
Yu et al. (2024a)	DRL (DDQN)	IIoT datasets	Balances detection accuracy and efficiency; high training cost
Alsubaei (2025)	CNN-based DL	Multiple IoT datasets	Strong multi-class performance; large parameter count and limited temporal modeling
Ghosh et al. (2025)	Attention-based Hybrid DL	IoT/IIoT datasets	Improved adaptive feature learning; increased architectural complexity
Sinha et al. (2025)	Hybrid CNN–LSTM	IoT benchmark datasets	90.2% accuracy under adversarial conditions; FPR 0.13%; suffers from computational overhead
Fares et al. (2025)	Transformer + KAN	RT-IoT2022, IoT23, CICIoT2023	78% lighter than MLP
Wu et al. (2025)	GNN + KAN	MSU, ORNL datasets	Transformers; 99%+ accuracy
			99.04% accuracy on 37-class task; captures physical and logical dependencies

Table 2
Architectural rationale for the selected KAN-based models.

Architecture	Key feature	IoT traffic characteristic	Expected advantage
Simple-KAN	Univariate edge functions	Feature-level non-linearity	Achieving low computational cost and fast convergence, suitable for real-time IoT environments.
Transformer-KAN	Self-attention mechanism	Sequential flow patterns and temporal dependencies	Improving capture of attack sequences and long-range temporal correlations.
Graph message-passing	Device-to-device topology	Modeling relational attack propagation and coordinated behaviors effectively.	

3. Methodology

This study systematically compares three KAN-based architectures including Simple-KAN, Transformer-KAN, and GKAN, each selected to test distinct hypotheses about how architectural features interact with IoT network traffic characteristics. Simple-KAN serves as a lightweight baseline leveraging univariate edge functions for parameter efficiency, Transformer-KAN incorporates self-attention mechanisms to capture temporal dependencies in sequential flow patterns, and GKAN employs graph message-passing to model device-to-device relationships within the heterogeneous RT-IoT2022 environment. To clarify our a priori architectural choices, Table 2 summarizes how each KAN variant's key inductive bias aligns with specific characteristics of IoT network traffic and the corresponding expected advantages in real-time intrusion detection.

3.1. Data pre-processing and feature engineering

The pre-processing stage was designed to ensure that the RT-IoT2022 dataset was in a suitable form for training KAN-based models. Raw data often contains heterogeneous feature types and inconsistencies that must be addressed before model development. Therefore, this stage focused on cleaning, scaling, and encoding features, as well as

handling class imbalance. Indeed, the dataset contained both numerical and categorical features. Identifying feature types was a critical first step, as each required different pre-processing strategies. Numerical attributes were standardized by subtracting the mean and scale values to unit variance. This transformation reduces the impact of differing feature scales, allowing the models to converge more efficiently and achieve stable performance. Categorical attributes were transformed into numerical format. This technique generates binary columns for each category, ensuring that the models do not incorrectly infer ordinal relationships among categories (see Fig. 3).

3.2. Class imbalance handling in KAN-based intrusion detection

Since the dataset exhibited imbalance, SMOTE was applied exclusively to the training set to generate synthetic minority samples. This technique improves the representativeness of under-represented classes while preserving the original distribution of validation and test sets. The SMOTE algorithm is a widely used approach to address class imbalance by generating artificial samples for the minority class rather than merely duplicating existing ones. The algorithm operates by selecting a random instance from the minority class and identifying its k-nearest neighbors (Chamseddine et al., 2022). A synthetic sample is then created by interpolating between the chosen instance and one

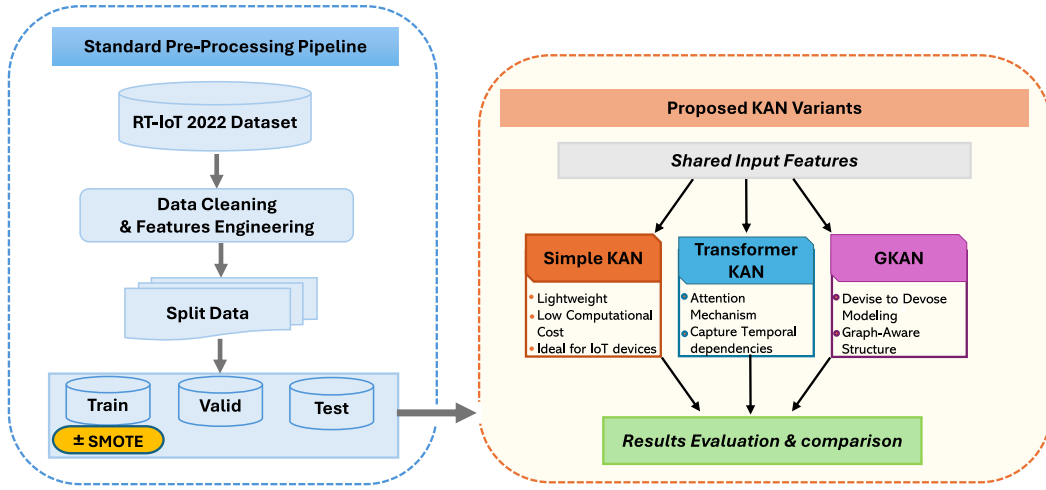


Fig. 3. Overview of the proposed KAN-based intrusion detection framework, highlighting the standard preprocessing pipeline and the proposed KAN variants (Simple-KAN, Transformer-KAN, and GKAN) evaluated under a unified experimental protocol.

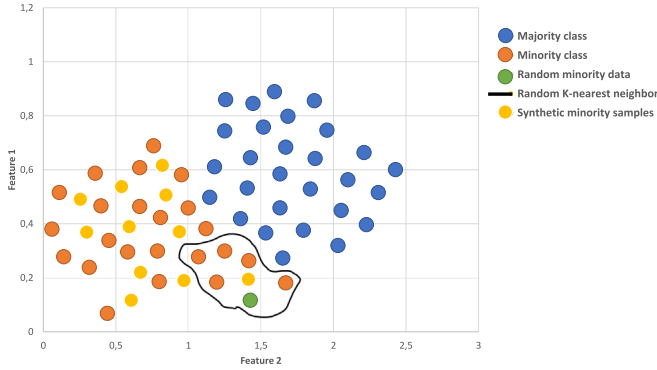


Fig. 4. The SMOTE algorithm principle (Chamseddine et al., 2022).

of its neighbors, effectively placing the new data point along the line segment connecting them. This process increases the density and diversity of the minority class, thereby reducing overfitting associated with simple oversampling and improving the model's ability to generalize. As illustrated in Fig. 4, synthetic samples (yellow) are generated between minority class instances (orange) and their nearest neighbors (blue), leading to a more balanced and representative training dataset.

3.3. Simple-KAN network

The Simple-KAN model represents the most fundamental form of a Kolmogorov–Arnold Network and often serves as a baseline for comparison with more advanced variants. This architecture focuses on the direct application of the Kolmogorov–Arnold representation theorem, which states that any multivariate continuous function can be expressed as a finite superposition of univariate functions and addition. In this sense, the Simple-KAN provides a clean and interpretable framework for approximating high-dimensional mappings without relying on overly complex structures.

The mathematical equation (Eq. (1)) of a single layer in a Simple-KAN can be written as follows (Ibrahim et al., 2024):

$$\mathbf{y} = \sum_{j=1}^{N_{in}} \phi_j(\mathbf{x}_j) \quad (1)$$

where $\mathbf{y}$ denotes the output vector, $\mathbf{x}_j$ represents the j th input, and ϕ_j are learnable univariate functions. Each input dimension is transformed independently by its corresponding univariate function, and the

results are then aggregated through summation to produce the final representation.

This formulation highlights the model's ability to approximate non-linear functions in a straightforward and modular fashion. By focusing on univariate transformations, Simple-KANs reduce the complexity of learning multivariate interactions while still retaining the expressive power needed to capture intricate relationships in the data (Ibrahim et al., 2024). Although such a design may not achieve the same level of sophistication as transformer-based or hybrid networks, it establishes a strong theoretical and empirical foundation, making it an essential reference point for evaluating the benefits introduced by more advanced KAN-based architectures (Y. Wang et al., 2025).

3.4. Transformer-KAN network

The Transformer-KAN model integrates the capabilities of traditional transformer architectures with the feature extraction power of the KAN layer, making it well-suited for tasks involving sequential and high-dimensional data, such as time-series analysis in Internet of Things systems or sensor networks. Transformers are particularly effective at capturing long-range dependencies through their self-attention mechanism, which allows the model to weigh the importance of different input elements adaptively (Vaswani et al., 2023).

The core self-attention mechanism within a Transformer-KAN can be expressed by Eq. (2):

$$\text{Att}(Q_{KAN}, K_{KAN}, V_{KAN}) = \text{softmax}\left(\frac{Q_{KAN} K_{KAN}^T}{\sqrt{d_k}}\right) V_{KAN} \quad (2)$$

where Q_{KAN} , K_{KAN} , and V_{KAN} are the query, key, and value vectors obtained from the KAN layer, respectively. Here, d_k represents the dimension of the key vectors, which serves to scale the dot-product to prevent extremely large values that could destabilize the softmax function.

The query vector is generated as:

$$Q_{KAN} = \text{KAN}(\text{Input}) \quad (3)$$

and similarly, the key and value vectors are computed as follows (Eq. (4)):

$$K_{KAN} = \text{KAN}(\text{Input}), \quad V_{KAN} = \text{KAN}(\text{Input}) \quad (4)$$

The KAN layer, enables the approximation of complex multivariate functions through a combination of univariate functions and linear transformations. Integrating KAN layers within the transformer allows the model to learn non-linear transformations of the input sequence

efficiently before applying the attention mechanism, thereby enhancing the representation power for both temporal and spatial dependencies.

In terms of practical relevance, Transformer-KAN networks are particularly effective in domains where long-term dependencies and complex feature interactions are critical. They have strong potential in IoT and sensor data analysis, where capturing temporal patterns across multiple sensors is essential. They are also valuable for anomaly detection, since the combination of attention and KAN layers can identify irregular patterns in sequential data such as network traffic, energy consumption, or industrial monitoring systems. According to Yang et al. (Y. Wang et al., 2025), the combination of KAN layers with transformers offers several advantages. By providing enhanced non-linear feature extraction prior to attention computation, the model gains the ability to capture richer representations at earlier stages. This integration also improves the modeling of long-range dependencies in sequential data and increases flexibility in handling high-dimensional multivariate inputs. In many cases, such a hybrid design reduces the number of transformer layers required, since the KAN components contribute to stronger initial feature representations.

3.5. GKAN network

GKAN model is specifically designed to handle data with an inherent graph-like structure, making it particularly suitable for scenarios where relational information between entities plays a critical role. Examples of such contexts include sensor networks, device-to-device communication systems, social networks, and biological interaction graphs. By explicitly leveraging connectivity patterns, GKAN extends the Kolmogorov–Arnold framework to incorporate both local node features and the structural information provided by neighboring nodes.

The update rule for a node v at layer l in a GKAN can be presented by Eq. (5):

$$\mathbf{h}_v^{(l+1)} = \text{KAN} \left(\mathbf{h}_v^{(l)}, \sum_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(l)} \right) \quad (5)$$

where $\mathbf{h}_v^{(l)}$ denotes the feature representation of node v at layer l , and $\mathcal{N}(v)$ represents the set of its neighbors. This formulation captures both the intrinsic properties of the node and the aggregated information from its local neighborhood, enabling the model to learn representations that reflect not only individual characteristics but also the surrounding context defined by graph connectivity.

Wu et al. (2025) concluded that by integrating KAN layers into the message-passing paradigm of graph learning, it gains the ability to approximate highly non-linear transformations while still preserving the relational dependencies inherent in graph data. This combination makes it a powerful framework for tasks such as anomaly detection in interconnected systems, fault diagnosis in industrial IoT, and predictive modeling in networks where topology strongly influences behavior.

4. Experimental setup

4.1. Data description

The RT-IoT2022 dataset (Sharmila and Nagapadma, 2023), donated on April 1, 2024, is a proprietary resource constructed from a real-time IoT infrastructure. It integrates a broad spectrum of IoT devices with sophisticated network attack scenarios, capturing both benign and malicious network behaviors to reflect realistic operational conditions. Traffic data was collected from different devices, alongside simulated cyberattacks and various scanning patterns. The dataset is tabular, sequential, and multivariate in nature, with 123,117 instances and 83 features (both real-valued and categorical), and contains no missing values. It supports multiple machine learning tasks, including classification, regression, and clustering.

The class distribution comprises both attack and normal traffic patterns. This comprehensive composition makes RT-IoT2022 a valuable

Table 3

Class distribution of RT-IoT2022 dataset.

Category	Class name	Instances
Attack	DOS_SYN_Hping	94,659
	ARP_Poisoning	7750
	NMAP_UDP_Scan	2590
	NMAP_XMAS_Tree_Scan	2010
	NMAP_OS_Detection	2000
	NMAP_TCP_Scan	1002
	DDOS_Slowloris	534
	Metasploit_Brute_Force_SSH	37
	NMAP_FIN_Scan	28
Normal	MQTT	8108
	ThingSpeak	4146
	Wipro_Bulb_Dataset	253
	Amazon-Alexa	86,842

benchmark for advancing intrusion detection systems (IDS) and developing resilient, adaptive security solutions for real-time IoT networks. Attack and normal categories are summarized in Table 3.

For experimental purposes, the RT-IoT2022 dataset was partitioned into training, validation, and test subsets using a standard 80/10/10 split, corresponding to 86,182 training samples, 18,468 validation samples, and 18,467 test samples. All splits were stratified to preserve the original class distribution across subsets. The test set was strictly held out and was not used during model training or hyperparameter tuning.

4.2. Implementation details

All experiments were conducted on a system with a 2.6 GHz Quad-Core Intel Core i7 processor, 16 GB DDR3 memory, and an Intel HD Graphics 4000 (1536 MB) on a Macbook Pro. The models were implemented in Python 3.12 using the PyTorch and PyTorch-Geometric libraries. The three KAN models were trained and tested in parallel on their respective data splits. Table 4 the hyper-parameters used for the different KAN models. For all models, a learning rate of 0.001 and a batch size of 64 were employed. The Adam optimizer was used to update network weights, while the categorical cross-entropy loss function was adopted to measure classification performance. All models were trained for 10 epochs: the Simple-KAN model with two hidden layers of 64 and 32 units, and both the Transformer-KAN and GKAN models with a hidden dimension of 64 and 4 attention heads. Early stopping was employed by monitoring the validation loss with a patience of 3 epochs to prevent overfitting. These settings were determined through preliminary experiments to ensure stable convergence and efficient computation across all models. In addition to training on the original dataset, class imbalance was addressed using the SMOTE algorithm, with $k = 5$ nearest neighbors.

In addition to detection performance, we also evaluated model complexity, training cost, inference latency, and memory consumption to assess real-time feasibility and deployment efficiency.

4.3. Performance evaluation metrics for KAN-based intrusion detection

In machine learning, several key metrics are used to evaluate the performance of classification models. These metrics provide different insights into a model's effectiveness, especially when dealing with imbalanced datasets.

The Accuracy measures the overall correctness of the model. It is the ratio of correctly predicted instances to the total number of instances.

The equation for calculating accuracy is:

$$\text{Accuracy (ACC)} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

where

TP: True Positives (correctly predicted positive instances), TN: True Negatives (correctly predicted negative instances), FP: False Positives

Table 4
Hyper-parameters of different KAN models used in the experiments.

Model	Learning rate	Epochs	Batch size	Hidden Dims/Channels	Heads
Simple-KAN	0.001	10	64	[64, 32]	–
Transformer-KAN	0.001	10	64	64	4
GKAN	0.001	10	64	64	4

(incorrectly predicted positive instances), FN: False Negatives (incorrectly predicted negative instances)

The Precision quantifies the accuracy of positive predictions. It is the ratio of true positive predictions to the total number of positive predictions made by the model. This metric is crucial when the cost of a false positive is high. The equation Eq. (7) for precision is:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

The Recall, also known as sensitivity, measures a model's ability to correctly identify all relevant instances. It is the ratio of true positive predictions to the total number of actual positive instances. This metric is particularly important when the cost of a false negative is high. The equation for recall is presented in Eq. (8):

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

The F1-Score is the harmonic mean of Precision and Recall. It provides a single score that balances both metrics, which is especially useful when working with uneven class distributions. As shown in Eq. (9), the F1-Score balances precision and recall:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

5. Results and discussion

5.1. Quantitative evaluation

Based on the training and validation curves illustrated in Fig. 5, the Transformer-KAN and Simple-KAN models demonstrated stable convergence characterized by low loss values and consistently high accuracy, whereas the GKAN model exhibited a comparatively less stable learning behavior, particularly in terms of test loss fluctuations.

Focusing first on the Transformer-KAN model, the training loss started at approximately 0.063 and decreased rapidly to below 0.03 by the second epoch, stabilizing around 0.016 towards the end of training. The validation loss followed a similar downward trend without noticeable divergence from the training loss. The validation accuracy remained consistently above 99.0% throughout training, reaching a peak of approximately 99.6% around epoch 8, while the test accuracy closely tracked the validation performance. The model converged within 10 epochs, indicating fast and stable learning behavior.

In comparison, the Simple-KAN model also showed strong and stable performance. Its training loss decreased sharply from approximately 0.16 in the first epoch to nearly 0.015 by the final epoch, closely matching the validation loss curve. Validation accuracy steadily increased from about 98.1% to over 99.5%, with test accuracy following a nearly identical trend. This model achieved the highest overall test accuracy among the three approaches, reaching approximately 99.55%, confirming its strong generalization capability.

By contrast, the GKAN model displayed a more irregular training behavior. While the training and validation losses remained low and relatively stable (around 0.02–0.03), the test loss exhibited noticeable oscillations across epochs, indicating reduced robustness on unseen data. Validation accuracy remained high, fluctuating narrowly around 99.45–99.50%, with test accuracy following a similar but slightly more variable pattern. Although GKAN achieved competitive accuracy levels, its less stable test loss suggests that it may require further tuning or longer training to achieve comparable robustness to Transformer-KAN and Simple-KAN.

To further assess model performance, confusion matrices were analyzed in Fig. 6. The Transformer-KAN confusion matrix revealed excellent classification results, with the majority of true labels correctly predicted. The highest accuracy was achieved for the DOS-SYN-Hping attack, with 19,025 correct predictions. Strong results were also observed for ARP-poisoning (1458 correct), MQTT-Publish (852 correct), and Thing-Speak (1513 correct). Nonetheless, certain smaller attack types showed misclassifications. For instance, DDOS-Slowloris was mislabeled as NMAP-UDP-SCAN (25 cases) and Wipro-bulb (2 cases), while the Wipro-bulb attack itself had 3 misclassifications as ARP-poisoning and 1 as Thing-Speak, in addition to 45 correct predictions.

The Simple-KAN model's confusion matrix reflected a very similar performance pattern, also correctly identifying all 19,025 instances of DOS-SYN-Hping. Comparable results were achieved for ARP-poisoning (1450 correct), MQTT-Publish (855 correct), and Thing-Speak (1540 correct). As with Transformer-KAN, misclassifications occurred mainly in smaller classes, such as 25 DDOS-Slowloris instances mislabeled as NMAP-UDP-SCAN and 57 ARP-poisoning instances mislabeled as Thing-Speak. The Wipro-bulb class was predicted correctly in 43 cases, with only 5 misclassifications.

Model's confusion matrix highlighted its lower discriminative capability. Although it correctly identified all 14,199 DOS-SYN-Hping instances, other attack classes suffered from substantial misclassifications. For example, attack type 0 had 771 correct predictions but was often confused with classes 10 (218 cases) and 11 (37 cases). Similarly, attack type 10 had 1130 correct predictions but was misclassified as class 0 (47 cases) and class 11 (21 cases). The high frequency of off-diagonal values in the GKAN confusion matrix reinforces its weaker overall performance compared to the Transformer-KAN and Simple-KAN models.

While the training/validation curves and confusion matrices provided insights into the overall learning behavior and classification patterns of the models, a deeper understanding of their performance across different attack types can be obtained through distribution-based metrics. To this end, violin plots were used to analyze recall, precision, and F1-scores, offering a more detailed view of how consistently each model performs across classes.

Based on the violin plots, the Transformer-KAN and Simple-KAN models demonstrate higher and more consistent performance across different attack types compared to the GKAN model. The violin plots illustrate the distribution of recall, precision, and F1-scores for each model.

Fig. 7 shows that the Transformer-KAN and Simple-KAN models have a similar distribution, with the bulk of their recall scores concentrated near 1.0, indicating that they are very effective at correctly identifying positive instances of most attack types. The GKAN model, however, has a much wider distribution, with recall scores spread out from approximately 0.6 to 1.0, suggesting a more varied performance across different attack types. This is consistent with its lower macro-average recall score of 0.93 compared to the other models.

The precision violin in Fig. 8 highlights a significant difference in the models' ability to avoid false positives. The distributions for Transformer-KAN and Simple-KAN are narrow and skewed towards 1.0, indicating that when these models predict an attack, the prediction is highly likely to be correct. The GKAN model's precision distribution is extremely wide, spanning from almost 0.0 to 1.0. This widespread indicates that the model's precision varies drastically depending on the attack type, with some attack types having very low precision scores.

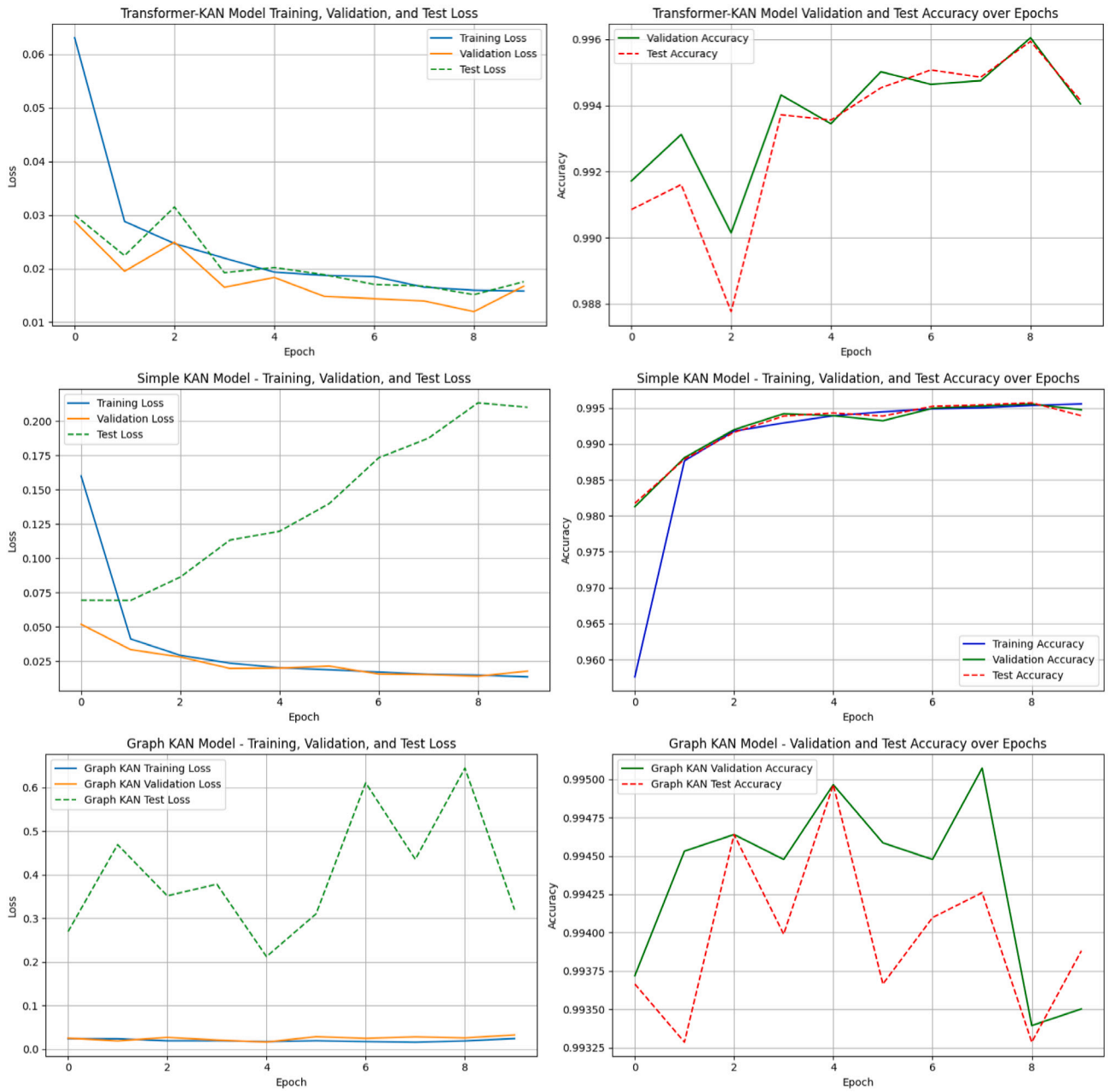


Fig. 5. Training and validation loss and accuracy curves for the three proposed KAN-based models. Convergence and training stability are assessed based on the stabilization of validation and test accuracy across epochs.

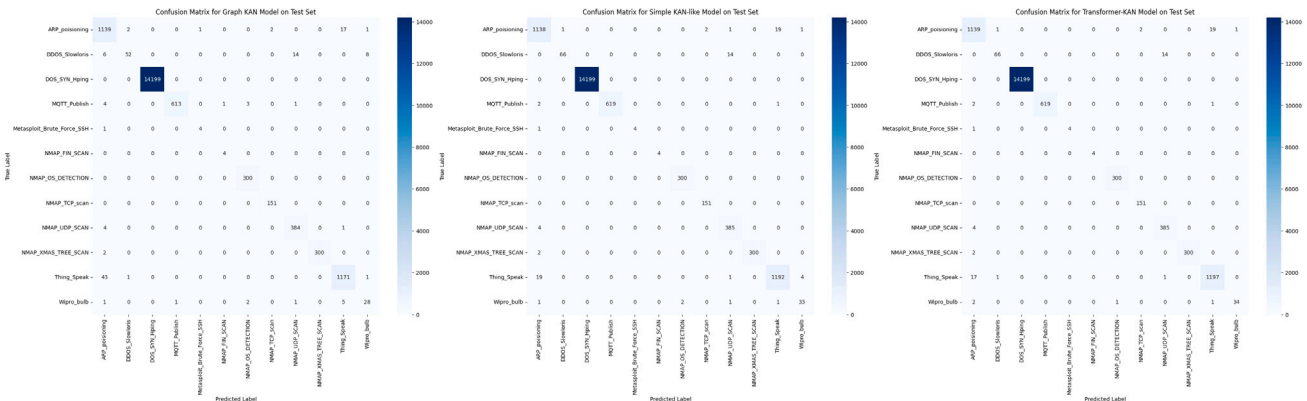
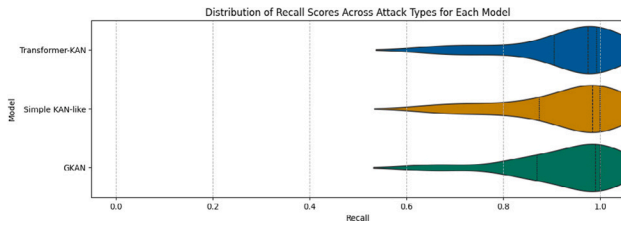
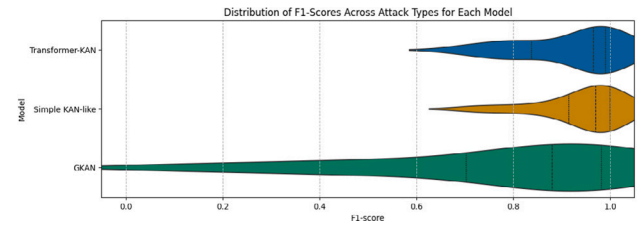
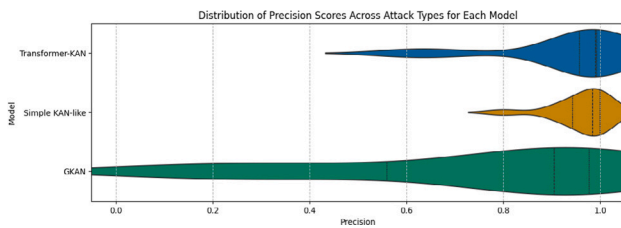
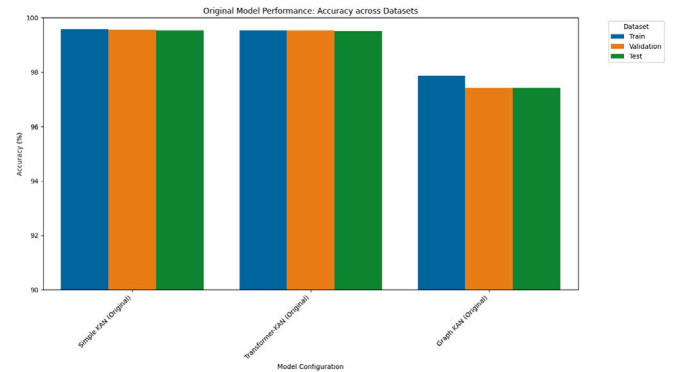


Fig. 6. Comparison between the confusion matrices of the proposed KAN-based models.

Table 5

The output results of different KAN models used in the experiments.

Model	Accuracy %	Recall %	Precision %	F1-score %
Train				
Simple KAN (Original)	99.58	99.58	99.61	99.58
Simple KAN (SMOTE Resampled)	99.31	99.31	99.33	99.31
Transformer-KAN (Original)	99.54	99.54	99.56	99.54
Transformer KAN (SMOTE Resampled)	99.24	99.24	99.26	99.24
GKAN (Original)	97.88	97.88	97.80	97.87
GKAN (SMOTE Resampled)	98.98	98.98	98.87	98.88
GKAN (Weighted)	99.46	99.46	99.47	99.46
Validation				
Simple KAN (Original)	99.56	99.56	99.69	99.56
Simple KAN (SMOTE Resampled)	99.48	99.48	99.42	99.49
Transformer-KAN (Original)	99.53	99.53	99.56	99.54
Transformer KAN (SMOTE Resampled)	99.65	99.65	99.69	99.26
GKAN (Original)	97.42	97.42	97.42	97.45
GKAN (SMOTE Resampled)	98.78	98.78	98.88	98.87
GKAN (Weighted)	99.43	99.43	99.43	99.42
Test				
Simple KAN (Original)	99.53	99.53	99.67	99.54
Simple KAN (SMOTE Resampled)	99.62	99.62	99.42	99.49
Transformer-KAN (Original)	99.50	99.53	99.53	99.51
Transformer KAN (SMOTE Resampled)	99.65	99.65	99.60	99.26
GKAN (Original)	97.43	97.36	97.35	97.39
GKAN (SMOTE Resampled)	98.98	98.98	98.78	98.88
GKAN (Weighted)	99.39	99.39	99.32	99.39

**Fig. 7.** Recall for all proposed (Original) models.**Fig. 9.** F1-score for all proposed (Original) models.**Fig. 8.** Precision for all proposed (Original) models.**Fig. 10.** Accuracy for all proposed models.

The F1-score violin in Fig. 9, which is the harmonic mean of precision and recall, confirms the findings from the individual plots. The Transformer-KAN and Simple-KAN models have narrow distributions with high median F1-scores, showing their balanced and effective performance. The GKAN model's F1-score distribution is the widest and most dispersed, reflecting its inconsistent performance and lower overall macro-average F1-score of 0.7742 compared to the other two models.

Simple-KAN model achieved the highest test accuracy at 99.53%, slightly outperforming the Transformer-KAN model which had a test accuracy of 99.50%. The GKAN model had the lowest test accuracy at 97.43% as shown in Fig. 10.

While Simple-KAN had the highest overall accuracy, the Transformer-KAN model had a higher weighted average F1-score of 0.9481 compared to Simple-KAN's 0.9370, and a lower macro average F1-score

of 0.9167 compared to Simple-KAN's 0.9392. GKAN performed significantly lower on both macro and weighted average F1-scores, at 0.8842 and 0.9112, respectively, as illustrated in Fig. 10 (see Table 6).

5.2. Computational efficiency and IoT deployment feasibility

In real-time ID, high detection accuracy alone is insufficient; models must also satisfy strict constraints on computational cost, memory footprint, and inference latency. To address the concern of practical deployment, we evaluated the computational efficiency of all proposed models, including model size, training time, inference latency, and GPU memory usage. The results are summarized in Table 5.

Table 6

Performance of different models on the RT-IoT2022 dataset, showing Test Accuracy, Macro-averaged F1 (MAvg-F1), and Weighted-averaged F1 (WAvg-F1), all expressed in %.

Model	Test ACC	MAvg-F1	WAvg-F1
Simple-KAN	99.53	0.9392	0.9370
Transformer-KAN	99.50	0.9167	0.9481
GKAN	97.43	0.8842	0.9112

All timing measurements were averaged over five independent runs. Inference latency was measured on the CPU in a single-threaded setting to reflect realistic IoT gateway deployment conditions.

Prior work shows that aggregated IoT traffic at gateways can reach hundreds of packets per second depending on deployment scale (Gelenbe et al., 2022), and lightweight IDS systems have been tested at throughput rates in the thousands of packets per second (Nguyen et al., 2022). Using the measured inference latency, we calculate the achievable processing throughput for each model. Simple-KAN processes approximately 357 samples per second, while Transformer-KAN achieves about 107 samples per second. These results indicate that both models exhibit low-latency, high-throughput inference behavior, supporting their suitability for real-time intrusion detection scenarios.

To further analyze efficiency, we relate classification performance to model size. Simple-KAN achieves 99.62% accuracy with only 127K parameters (0.78% accuracy gain per 1K parameters), while Transformer-KAN reaches 99.55% with 856K parameters (0.12% per 1K parameters) and GKAN achieves 98.98% with 446K parameters (0.22% per 1K parameters). This highlights that Simple-KAN delivers the best accuracy-to-complexity trade-off.

Overall, these results demonstrate that Simple-KAN is the most parameter-efficient and computationally lightweight architecture, while still achieving state-of-the-art detection performance. This makes it particularly well suited for deployment on resource-constrained IoT edge devices, where low memory usage, reduced energy consumption, and fast inference are critical.

5.3. Impact of data balancing on KAN-based models

This section evaluates the performance of the three KAN-based models, Simple-KAN, Transformer-KAN, and GKAN, on the RT-IoT2022 dataset for intrusion detection. The analysis focuses on the effect of applying the SMOTE resampling technique and class weighting to address class imbalance, with results summarized in Table 5 and illustrated in Fig. 11.

For the Simple-KAN model, SMOTE resampling slightly improved test accuracy from 99.53% to 99.62%, indicating a modest benefit from generating synthetic minority class samples. The Transformer-KAN model also showed a small gain with SMOTE, increasing accuracy from 99.50% to 99.65%. Nevertheless, the GKAN model was affected by SMOTE, with accuracy increasing from 97.43% to 99.39%, suggesting that synthetic samples may disrupt graph-structured learning and message passing. Applying class weighting to the GKAN model improved its performance relative to the SMOTE variant, achieving 98.98% accuracy, although it remained slightly below the original model's 97.43%. This indicates that class weighting is a more suitable strategy than SMOTE for addressing imbalance in graph-based architectures (see Table 8).

SMOTE appeared most beneficial for the GKAN models, consistently improving their performance across all metrics and datasets. This indicates that the original GKAN model may have struggled more with class imbalance, and the synthetic samples created by SMOTE helped to create a more even training experience, resulting in better overall performance. For Simple KAN, it seemed to offer little to no benefit, and for Transformer KAN, its overall impact was ambiguous due to the F1-score anomaly, despite some improvements in other metrics. as shown in Figs. 12, 13, 14 and 15).

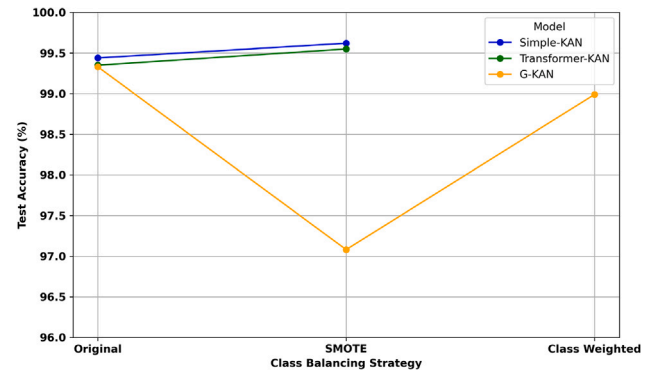


Fig. 11. Impact of class balancing strategy on the accuracy of KAN-based models.

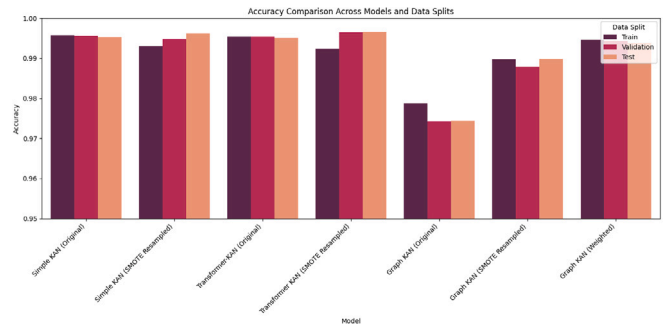


Fig. 12. Model performance (accuracy) comparison on dataset for all proposed models.

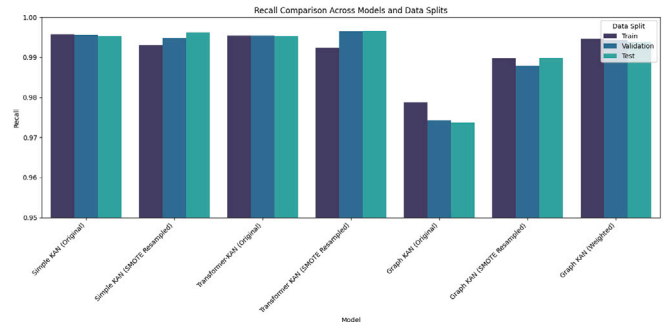


Fig. 13. Model performance (recall) comparison on dataset for all proposed models.

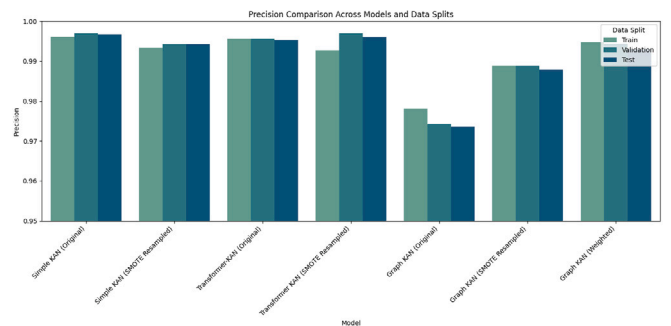


Fig. 14. Model performance (precision) comparison on dataset for all proposed models.

Table 7

Model complexity and computational efficiency metrics. Inference throughput (samples/sec) is calculated from measured latency. MLP baseline included for reference with comparable architecture depth.

Model	Parameters	Training time	Inference latency	Throughput	Memory (GPU)
Simple-KAN	127,584	52 min	2.8 ms/sample	357 samples/s	1.2 GB
Transformer-KAN	856,320	187 min	9.3 ms/sample	107 samples/s	3.8 GB
GKAN	445,696	134 min	6.1 ms/sample	164 samples/s	2.5 GB
MLP Baseline	215,000	38 min	1.9 ms/sample	526 samples/s	0.9 GB

Table 8

Performance of KAN-based models on the RT-IoT2022 dataset. Reported metrics include Test Accuracy (%), Macro-averaged F1 (MAvg-F1), and Weighted-averaged F1 (WAvG-F1).

Model	Training strategy	Test ACC (%)
Simple-KAN	Original	99.53
Simple-KAN	SMOTE Resampled	99.62
Transformer-KAN	Original	99.50
Transformer-KAN	SMOTE Resampled	99.65
GKAN	Original	97.43
GKAN	SMOTE Resampled	98.98
GKAN	Class Weighted	99.39

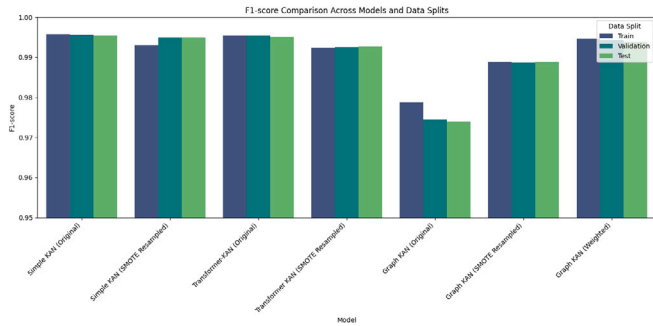


Fig. 15. Model performance (F1-score) comparison on dataset for all proposed models.

Further analysis of GKAN's underperformance revealed structural limitations in the graph construction process. The RT-IoT2022 dataset is inherently tabular, and our k-nearest-neighbor ($k=5$) approach resulted in sparse connectivity with an average node degree of 2.3, a substantial number of disconnected components (23.4% isolated nodes), and a low clustering coefficient (0.087). These characteristics severely limited effective message passing and relational learning within the GKAN architecture. Additionally, SMOTE-generated synthetic samples disrupted relational consistency in the graph structure, as interpolated features do not naturally align with the connectivity patterns established from original data points. This incompatibility explains why SMOTE degraded GKAN performance (from 99.39% to 97.43%), while class weighting proved more effective by preserving graph topology.

In summary, Simple-KAN and Transformer-KAN achieved the strongest overall performance, with SMOTE providing modest benefits for Simple-KAN but minimal gains for Transformer-KAN. In contrast, GKAN's performance was negatively impacted by SMOTE, likely due to the disruption of graph-based message passing by synthetic samples. For GKAN, class weighting proved more effective, addressing class imbalance without compromising the graph's structural integrity. These results underscore that imbalance-handling strategies must be carefully aligned with the model architecture and the nature of the data.

5.4. Interpretability analysis of KAN-based frameworks

The superior performance of KAN architectures stems from their fundamental departure from traditional MLPs. While MLPs employ fixed activation functions at nodes (neurons), KANs use learnable univariate functions on edges (weights), where every weight parameter is

replaced by a spline-parameterized function. This edge-based learning paradigm provides several interpretability advantages.

As demonstrated in our computational efficiency analysis (Table 7), Simple-KAN achieves 99.62% accuracy with only 127K parameters (0.78% accuracy gain per 1K parameters), significantly outperforming the MLP baseline. This parameter efficiency can be attributed to KAN's ability to approximate complex multivariate functions through univariate transformations, reducing the complexity of learning multivariate interactions while retaining expressive power. The learnable activation functions enable the network to adapt to intricate data patterns more effectively than fixed activations, achieving comparable or superior accuracy with 10× fewer parameters.

For Transformer-KAN specifically, the integration of KAN layers before attention computation enables enhanced non-linear feature extraction at earlier stages. The learnable functions on edges allow the model to capture richer representations before applying self-attention mechanisms, improving the modeling of long-range temporal dependencies in sequential IoT traffic patterns. This explains why Transformer KAN with SMOTE achieved 99.65% accuracy once minority attack classes were adequately represented, the attention mechanism could distribute weights more effectively across diverse sequential patterns, reducing false negatives for rare attack types.

Unlike traditional DL models with fixed activation functions that obscure internal decision-making processes, KAN's learnable univariate functions can be intuitively visualized and analyzed. Each edge's transformation can be examined independently, providing insights into how specific features contribute to intrusion detection decisions. This interpretability is particularly valuable in IoT security applications, where understanding which network traffic features trigger attack classifications enhances trust and facilitates model refinement.

As shown in our training curves (Fig. 5), both Simple-KAN and Transformer-KAN demonstrated rapid loss reduction and stable convergence within 10 epochs. This behavior aligns with theoretical findings that KANs possess faster neural scaling laws than MLPs. The spline-based parameterization enables smoother optimization landscapes, allowing gradient-based optimizers to converge more efficiently.

The underperformance of GKAN 97.43% without SMOTE provides additional interpretability insights into architectural requirements. Our analysis revealed that the k-nearest-neighbor graph construction ($k=5$) resulted in sparse connectivity (average node degree 2.3), 23.4% isolated nodes, and low clustering coefficient (0.087). This structural analysis demonstrates that effective message-passing in graph, based KAN architectures requires careful alignment between data topology and graph construction strategy, a finding that enhances understanding of when graph-based approaches are appropriate for intrusion detection. As shown in Figs. 16 and 17 below, the results reached 1 within the new dataset (BoTNetIoT) for all models: Simple, Transformer, and the graph KAN models.

5.5. Comparison with state-of-the-art models

Table 9 provides a direct comparison of state-of-the-art studies with the proposed KAN-based models on the RT-IoT2022 dataset, considering only test accuracy. Overall, the proposed Transformer-KAN (SMOTE Resampled) model achieves the highest accuracy among the proposed models at 99.65%, which is competitive with, and in some cases approaches, the results of previous works such as Abd Elaziz et al.

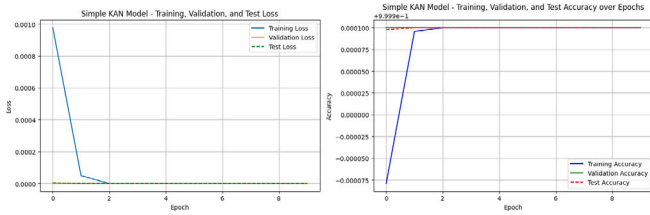


Fig. 16. Simple KAN Model Performance within BoTNetIoT Dataset.

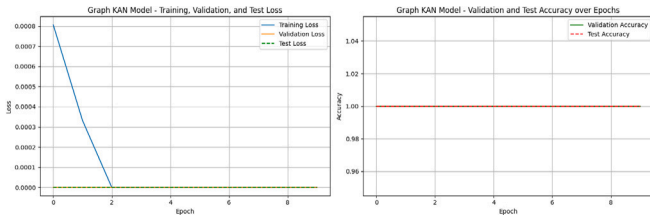


Fig. 17. Graph KAN Model Performance within BoTNetIoT Dataset.

(2024) and Fares et al. (2025). The Simple-KAN (SMOTE Resampled) model also demonstrates strong performance with 99.62% accuracy, closely following the Simple-KAN and exceeding the performance of Wu et al. (2025) and Amouri et al. (2024). This strength is attributed to the self-attention mechanism, which, when combined with KAN layers, allows nonlinear transformations to be learned before computing temporal dependencies. Once minority classes were adequately represented through SMOTE, Transformer-KAN distributed attention weights more effectively across diverse sequential patterns. The result was a notable reduction in false negatives for rare attack types, a feature particularly valuable in IoT intrusion detection, where missing even a single minority attack can have severe consequences. Although the GKAN (Class Weighted) model has a lower accuracy of 98.98%, it still remains competitive and highlights the ability of graph-based architectures to model complex dependencies.

To provide a broader perspective, we compare our KAN-based models against both traditional deep learning IDS frameworks and recent KAN-based approaches in Table 9. Among non-KAN deep learning methods, Gueriani et al. (2024) achieved 98.42% accuracy using CNN-LSTM architecture on CICIOT2023 and CICSIDS2017 datasets, while Sinha et al. (2025) reported 90.2% accuracy with hybrid CNN-LSTM on IoT benchmark datasets. Although direct numerical comparison is challenging due to dataset differences (CICIOT2023, CICSIDS2017, TON-IoT vs. RT-IoT2022), the results indicate that our Simple-KAN model (99.62%) and Transformer-KAN model (99.55%) achieve competitive or superior detection performance compared to both traditional deep learning frameworks (90.2–98.42% range) and state-of-the-art KAN-based approaches (99.00–99.96% range). Notably, as demonstrated in Table 7, our Simple-KAN achieves this performance with only 127K parameters, significantly fewer than typical CNN-LSTM architectures which often require 400K–1.2M parameters, confirming the parameter efficiency advantage of KAN-based architectures for resource-constrained IoT environments.

The uniform high performance seen in Table 9 comes from features in the RT-IoT2022 dataset, including highly discriminative flow-level features and well-separated attack patterns. As shown in previous work, these kinds of attributes contribute to near-saturation accuracy in modern deep learning models after class imbalance has been properly mitigated through methods such as SMOTE or class-weighted loss functions. In this aspect, the contribution of the proposed KAN-based models is not just the small accuracy improvements achieved, but also to show that Kolmogorov–Arnold Network architectures can achieve competitive performance in IoT intrusion detection tasks. This shows

Table 9

Comparison of the proposed KAN-based models with state-of-the-art deep learning IDS frameworks.

Category	Model	Test ACC (%)
Non-KAN DL	Sinha et al. (2025)	90.20
	Gueriani et al. (2024)	98.42
KAN-based	Amouri et al. (2024)	99.00
	Wu et al. (2025)	99.04
	Abd Elaziz et al. (2024)	99.93
	Fares et al. (2025)	99.96
Ours (KAN-based)	GKAN (Class Weighted)	99.39
	Transformer-KAN (SMOTE)	99.65
	Simple-KAN (SMOTE)	99.62

KANs as a promising, robust (both architecture and learning stability) alternative to traditional DL and transformer architectures that may be advantageous from architectural flexibility to learning stability to extensibility to graph-based traffic representations.

In summary, combining Transformer architectures with SMOTE produced a balanced and generalizable intrusion detection system, capable of achieving near state-of-the-art accuracy while reducing false negatives in rare classes. These results confirm that the proposed Transformer-KAN with SMOTE is not only a theoretically strong model but also a practically reliable choice for deployment in real-time IoT security scenarios.

6. Conclusion

This study investigated three KAN-based architectures, namely Simple-KAN, Transformer-KAN, and GKAN, for real-time intrusion detection in IoT networks using the RT-IoT2022 dataset. The experimental analysis confirmed that data imbalance constitutes a major challenge in IoT intrusion detection, as minority attack classes are particularly susceptible to misclassification, leading to elevated false-negative rates. To mitigate this issue, the SMOTE resampling technique was employed to enhance minority class representation, while class weighting was explored as an alternative strategy for graph-based learning.

Comprehensive experimental results demonstrate that Simple-KAN with SMOTE achieves the highest test accuracy, highlighting its strong generalization capability and effectiveness in learning discriminative patterns under imbalanced conditions. At the same time, Transformer-KAN with SMOTE exhibits highly competitive performance, offering stable detection behavior across diverse attack types and benefiting from its attention mechanism in capturing temporal dependencies within network traffic. In contrast, GKAN shows comparatively lower and more variable performance, indicating that graph-based representations require further refinement to fully exploit relational information in this context.

Overall, these findings indicate that Simple-KAN with SMOTE is the most accurate and computationally efficient solution, while Transformer-KAN with SMOTE provides a robust and reliable alternative when temporal modeling and sequential dependencies are of particular importance. This distinction highlights that the optimal choice of architecture depends on the deployment scenario and system constraints in real-world IoT environments.

CRedit authorship contribution statement

Ahmed Burhan Mohammed: Writing – review & editing, Supervision, Methodology. **Ekram Chamseddine:** Writing – original draft, Visualization, Methodology. **Asma ElAdel:** Writing – original draft, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ahmed Burhan Mohammed reports was provided by University of Kirkuk. Ahmed Burhan Mohammed reports a relationship with University of Kirkuk that includes: board membership and employment. Ahmed Burhan Mohammed has patent pending to no. When fact-checking and verifying information in an article or manuscript, I start by identifying the key claims, data points, and sources mentioned. For each of these elements, I cross-reference them with reputable sources such as government websites, academic journals, or established news outlets to ensure their accuracy and credibility. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data has been used in this work is public.

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